READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 10

Derivatives: Pricing, Strategies and Risk Management (Part 2)

STUDY NOTES · EDITION 1.0 · WEEK 10

How to use these notes

These notes cover the whole of Week 10: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A introduces the futures contract — the opposite of the option built in Week 9 — and compares the two directly. Part B covers the machinery that makes futures work: margin, daily settlement, leverage and hedging. Part C returns to options and combines them into four strategies. Read the parts in order; the Zomato example from Week 9 continues throughout.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

Contents

Module overview

Week 9 built the option: a right that can be abandoned for the cost of a premium. This module introduces its opposite, the futures contract, where both parties are bound and neither pays the other anything at the outset. It then covers the machinery that makes such promises enforceable — margin and daily settlement — and the leverage that machinery creates. It closes by combining shares, calls and puts into four strategies.

One thread runs through all nine topics: **a right must be purchased, but a promise exchanged for another promise costs nothing**. That single asymmetry explains the premium, the payoff shape, the margin requirement and the risk profile of every instrument in the week.

Learning objectives

By the end of this module, you will be able to:

1. **Explain** how a forward becomes a futures contract, and compare futures with options.
2. **Describe** margin and daily settlement, and compute leverage and its effect.
3. **Compute** the number of index futures contracts needed to hedge a portfolio.
4. **Apply** moneyness and construct the protective put, covered call, straddle and strangle.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	From forward to futures	A · The promise	L35 V1–V2	LO 1
2	The futures payoff and index futures	A · The promise	L35 V3	LO 1
3	Options versus futures	A · The promise	L35 V4	LO 1
4	Margin and marking to market	B · Machinery	L35 V5	LO 2
5	Leverage	B · Machinery	L35 V5	LO 2
6	Hedging a portfolio with index futures	B · Hedging	L35 V6	LO 3
7	Moneyness	C · Strategies	L36 V1	LO 4
8	Protective put and covered call	C · Strategies	L36 V2	LO 4
9	Straddle and strangle	C · Strategies	L36 V3	LO 4

Key terms at a glance

Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Forward contract
A private, customised agreement to trade at a set price on a future date.

Futures contract
A standardised forward supported and guaranteed by an exchange.

Counterparty risk
The possibility that the other party fails to honour its promise.

Basis
The gap between spot and futures price, which converges to zero at expiry.

Long position
The side promising to buy; gains when the underlying rises.

Short position
The side promising to sell; gains when the underlying falls.

Margin
A refundable security deposit required of both parties to a futures contract.

Marking to market
Daily settlement of gains and losses into the margin account.

Margin call
A demand for additional funds when the balance falls below the required minimum.

Leverage
Contract value relative to margin deposited; magnifies gains and losses alike.

Index futures
Futures on the level of an index, settled in cash rather than by delivery.

Hedge ratio
The number of contracts needed to hedge a portfolio, adjusted for beta.

Moneyness
The relationship between spot and strike that determines intrinsic value.

In the money
Exercising immediately would produce a gain.

At the money
Spot equals strike; the premium is entirely time value.

Out of the money
Exercising immediately would not make sense.

Protective put
Owning shares and buying a put; creates a floor under losses.

Covered call
Owning shares and selling a call; creates a ceiling on gains.

Synthetic position
A payoff engineered from other instruments to imitate a different one.

Long straddle
A call and a put at the same at-the-money strike; a bet on size, not direction.

Long strangle
A call and a put at different out-of-the-money strikes; cheaper, needs a bigger move.

Financial engineering
Creating a desired cash flow pattern by combining financial instruments.

PART A · THE BINDING PROMISE

The instrument you cannot walk away from

1 From Forward to Futures

PART A · THE BINDING PROMISE · TOPIC 1 OF 9 · SERVES LEARNING OBJECTIVE 1

IN ONE LINE

A right must be purchased. A promise exchanged for another promise costs nothing at the outset — and binds both parties absolutely.

Why this matters

Week 9 built the option: a right that can be abandoned. This week opens with its opposite, and every difference between the two instruments flows from that single distinction.

When oil traded below zero

On 20 April 2020, the May West Texas Intermediate crude oil futures contract fell below zero and closed at minus $37.63 per barrel. Global lockdown had destroyed demand and storage at Cushing, Oklahoma was nearly full.

Why could oil have a negative price? Because many holders were traders, not oil companies. The contract required **physical delivery** of 1,000 barrels, and with storage almost full those traders had to either arrange costly storage or pay someone to take the obligation away.

Rahul's call option capped his maximum loss at ₹12,000. The oil traders held a binding promise. That is the contrast the whole week rests on.

Priya's second flat

Last week Priya paid ₹2 lakh for the right to buy a flat at ₹85 lakh. For a second flat the builder changes the terms: Priya pays **nothing** today, but after six months she must buy at ₹85 lakh and the builder must sell at the same price.

If the flat rises to ₹95 lakh she gains ₹10 lakh. If it falls to ₹70 lakh she must still pay ₹85 lakh and loses ₹15 lakh — her loss is not capped, because she cannot walk away.

The agreement is **symmetrical**. When Priya gains the builder loses, and vice versa. Neither side receives a one-sided right, which is exactly why no upfront premium is required.

The forward, and what an exchange adds

Ramu grows wheat near Karnal, ready for harvest in three months, selling at ₹25 per kilogramme. He fears a fall. Neha runs a flour mill and has contracted to supply a biscuit manufacturer at a fixed price; she fears a rise. Their fears are mirror images, so they agree that in three months Ramu will deliver at exactly ₹25 per kilogramme. No money changes hands — the agreement consists only of two promises.

Two problems follow. **Counterparty risk**: if wheat falls to ₹20, Neha would rather buy from the market, and Ramu's only recourse is a slow, expensive court. **Customisation**: the contract's exact quantity, quality, location and date may suit nobody else, so it cannot easily be transferred.

An exchange solves both. **Standardisation** fixes quantity, quality and delivery dates. A **central guarantee** means Ramu promises the exchange and the exchange promises Neha, so neither needs to trust a stranger. **A forward contract that has been standardised and supported by an exchange is a futures contract.**

TWO PROMISES, OR ONE RIGHT · THE DIFFERENCE THAT DEFINES THE WEEK

	OPTION · A RIGHT	FUTURES · A PROMISE
WHAT IT GRANTS	A right the buyer may use or abandon	A binding obligation on both parties
UPFRONT PAYMENT	A non-refundable premium	None — a refundable margin deposit instead
BUYER'S MAXIMUM LOSS	Limited to the premium — the payoff has a floor	No floor; losses grow as the market moves against you
SYMMETRY	One-sided — the buyer holds the right, the writer the obligation	Symmetrical — which is why nobody pays anybody

*The party promising to **buy** holds the long position; the party promising to **sell** holds the short. The long gains when prices rise; the short gains when they fall.*

Figure 1.1 A right must be bought. Two promises exchanged cost nothing — and bind absolutely.

KEY IDEA

The absence of a premium is not a discount. It is the direct consequence of symmetry: neither party has received anything the other has not, so there is nothing to pay for. And with nothing paid, there is nothing to walk away from.

EXAM POINTER

Define a forward, name both its problems — counterparty risk and customisation — and state precisely how an exchange solves each. The definition to memorise: a futures contract is a standardised forward supported and guaranteed by an exchange.

QUICK RECAP · TOPIC 1

- An option grants a right; a futures contract binds both parties absolutely.
- No premium is paid because the agreement is symmetrical.
- A forward is a private, customised agreement with counterparty risk and no easy exit.
- Standardisation and a central guarantee turn a forward into a futures contract.
- The long side promises to buy; the short side promises to sell.

2 The Futures Payoff and Index Futures

PART A · THE BINDING PROMISE · TOPIC 2 OF 9 · SERVES LEARNING OBJECTIVE 1

IN ONE LINE

A straight line with no kink, no floor and a break-even at exactly the locked price.

Why this matters

Compare this payoff with Week 9's hockey stick and the difference between a right and a promise becomes visible rather than abstract.

Rahul's futures trade

Rahul remains optimistic about Zomato at a spot price of ₹260. This time, instead of buying a right, he makes a binding promise: he agrees to purchase one Zomato futures contract at a futures price of **₹262** per share, lot size 1,000, expiring on the last Tuesday of the month. He pays no premium.

How the futures price is set

A convenient way to understand the futures price is as the expected future value of the underlying on the expiry date, assumed to grow at the risk-free rate. With a spot price of ₹250, an annual risk-free rate of 6% and one month to expiry: the monthly rate is 0.5%, so the theoretical futures price is 250 × 1.005 = **₹251.25**.

The difference between spot and futures price is the **basis**. During the contract's life the two may move differently, but as expiry approaches the gap narrows, and on the expiry date the basis becomes zero and the two prices converge. The basis is generally negative, because the futures price usually sits above spot due to the time value of money.

Three things the payoff shows

The break-even is simply ₹262, the locked futures price. There is nothing to add, because no premium was paid. Compare Week 9's call, where break-even was the ₹270 strike plus the ₹12 premium.

There is no kink. An option payoff bends at the strike because the buyer can choose not to exercise. A futures contract provides no right that can be abandoned, so nothing causes the line to change direction.

The loss is not capped. If Zomato fell to zero, Rahul would still be obligated to buy at ₹262 per share, losing ₹2,62,000. This is the same binding exposure that produced negative oil prices in 2020.

Index futures

The most heavily traded futures in India are often based on indices — Nifty 50 and Bank Nifty among them. The underlying is not a share but the numerical level of the index, so nobody can take delivery. These contracts are **settled in cash**: on expiry the difference between the locked level and the closing index level is paid, and no shares change hands.

THE FUTURES PAYOFF · LONG AT ₹262, LOT 1,000

Price on expiry	Long futures result	Total	Note
₹0 (worst case)	−₹262 per share	−₹2,62,000	Still obligated to buy at ₹262
₹252	−₹10 per share	−₹10,000	No floor — the line keeps falling
₹262 (break-even)	Nil	Nil	The locked price, with nothing to add
₹272	+₹10 per share	+₹10,000	One continuous straight line

*A short index futures at 24,500 with a lot of 25 units gains **₹12,500** if the index closes at 24,000 and loses ₹12,500 if it closes at 25,000. Break-even is exactly 24,500.*

Figure 2.1 The futures payoff. No kink, no floor, break-even at the locked price.

COMMON PITFALL

Describing a futures position as "buying at a discount". Nothing is bought at the outset and no premium is paid. Two promises are exchanged, which is precisely why neither side pays the other anything on day one.

EXAM POINTER

Give the theoretical futures price with the working, define basis and state that it converges to zero at expiry. For the payoff, name all three features — break-even at the locked price, no kink, no floor — and contrast each explicitly with the option payoff from Week 9.

QUICK RECAP · TOPIC 2

- Theoretical futures price = spot × (1 + risk-free rate for the period); ₹250 × 1.005 = ₹251.25.
- The basis is the gap between spot and futures, converging to zero at expiry.
- Break-even is the locked futures price, with no premium to add.
- The payoff has no kink, because there is no right to abandon, and no floor under the loss.
- Index futures are cash-settled, since nobody can take delivery of an index.

3 Options Versus Futures

PART A · THE BINDING PROMISE · TOPIC 3 OF 9 · SERVES LEARNING OBJECTIVE 1

IN ONE LINE

Insurance costs a premium but preserves the upside. A price lock is free but removes both outcomes. Neither is universally better.

Why this matters

This comparison is the most examinable content in the week, and the practical distinction at the end is what actually determines which instrument a desk chooses.

Seven differences

The comparison in the figure is worth learning as a block. Every row traces back to one thing: whether the instrument grants a right or imposes an obligation.

The practical distinction

A **protective put is insurance**. It costs a premium, but the holder keeps the entire upside. If the share rises, the put expires worthless and the gain is retained in full.

Ramu's forward cost nothing but was a **price lock**. It protected him from a fall and equally prevented him benefiting from a rise. If wheat had gone to ₹30, he would still have sold at ₹25.

Insurance requires payment but preserves the favourable outcome. A price lock is free but removes both outcomes. **There is no free protection**, and the investor must choose which risk to retain.

When each is used

A manager chooses **futures** when a low-cost, temporary, nearly complete hedge is needed — the objective is certainty and the upside is not the priority. A manager chooses **options** when protection must preserve the possibility of gains, and is willing to pay a premium for that.

SEVEN ROWS, ONE UNDERLYING DISTINCTION

	Option	Futures
What it grants	A right the buyer may use or abandon	A binding promise on both parties
Prices available	A range of strike prices, each a separate contract	One futures price for a given expiry
Upfront payment	A non-refundable premium, like insurance	No premium; a refundable margin deposit
Buyer's maximum loss	Limited to the premium — the payoff has a floor	No floor; losses grow with the market
Payoff shape	Bends at the strike — the hockey stick	One straight line through the locked price
Break-even	Strike + premium for a call; strike − premium for a put	Simply the locked futures price
Daily cash flows	Generally none until closed or expired	Marked to market every trading day

Insurance requires payment but preserves the favourable outcome. A price lock is free but removes both outcomes. *There is no free protection.*

Figure 3.1 The full comparison. Every row follows from right against obligation.

KEY IDEA

The choice is not about which instrument is safer. It is about *which risk you want to keep*. Futures remove the downside and the upside together; options remove the downside and charge you for keeping the upside.

EXAM POINTER

Comparison questions want at least five of the seven rows. The two that most often separate answers are the payoff shape — kink against straight line — and daily cash flows, since only futures are marked to market. Close with the insurance-versus-price-lock sentence.

QUICK RECAP · TOPIC 3

- An option grants a right; a futures contract imposes an obligation on both parties.
- Options cost a non-refundable premium; futures require a refundable margin deposit.
- The option payoff kinks at the strike; the futures payoff is one straight line.
- Futures are marked to market daily; options generally are not.
- Insurance preserves the upside at a cost; a price lock is free but removes both outcomes.

PART B · MARGIN, LEVERAGE AND HEDGING

What makes a promise enforceable

4 Margin and Marking to Market

PART B · MARGIN, LEVERAGE AND HEDGING · TOPIC 4 OF 9 · SERVES LEARNING OBJECTIVE 2

IN ONE LINE

A refundable deposit, settled every single evening — and the moment the balance runs short, the position may be closed for you.

Why this matters

Margin is what makes a futures exchange possible at all. It is also the mechanism that turns a temporary adverse move into a forced exit, which is the risk students most often overlook.

Why margin exists

Rahul controls Zomato shares worth ₹2,62,000 through his contract. Is it free? He does not know who is on the other side — he sees only his terminal, and the exchange stands between them guaranteeing both sides. That creates risk for the exchange: if Rahul cannot pay on expiry, someone must bear the consequences.

So both buyer and seller deposit money before entering. Contract value = ₹262 × 1,000 = ₹2,62,000. At an initial margin of 15.27%, that is approximately **₹40,000**, deposited by *both* parties.

What margin is not: it is not the price of the contract and not a fee. It is a refundable security deposit, like the deposit paid when renting a house.

Marking to market

Every evening after the close, the exchange settles the day's profit or loss rather than waiting until expiry. The day's gain is credited to the winning trader and the day's loss debited from the loser's margin account. This is **daily settlement**, or marking to market.

The margin call

With a minimum required balance of ₹30,000, watch what happens on day three. Rahul's balance falls to ₹28,000 — short of the threshold. His broker asks him to deposit more, and if he does not restore the balance the broker may close his position immediately.

That demand is a **margin call**, and it is one of the most feared communications in financial markets — not because of the amount, but because failing to meet it means being closed out at whatever price the market offers, regardless of whether the original view later proves right.

FOUR EVENINGS IN A MARGIN ACCOUNT

Day	Futures price	Change per share	Effect on account	Margin balance
Start	₹262	—	Initial margin deposited	₹40,000
Day 1	₹267	+₹5	+₹5,000	₹45,000
Day 2	₹255	−₹12	−₹12,000	₹33,000
Day 3	₹250	−₹5	−₹5,000	**₹28,000 — margin call**
Day 4	₹258	+₹8	+₹8,000	₹36,000

*Note day four. The price recovered and the account was restored — but a trader who could not meet the day three call would **already have been closed out** at ₹250.*

Figure 4.1 The daily settlement cycle. The call arrives before any recovery can help.

KEY IDEA

Marking to market means a futures loss is a **cash** loss the same evening, not a paper loss until expiry. That is the deepest difference from every instrument earlier in the course, and it is why solvency, not just judgement, determines who survives a position.

EXAM POINTER

Work the margin account day by day and identify exactly which day triggers the call. Define margin as a refundable deposit, not a fee or a price. The examinable insight is that a forced close can occur even when the eventual price move vindicates the position.

QUICK RECAP · TOPIC 4

- Margin is a refundable security deposit required of both parties, not a fee or a price.
- Contract value ₹2,62,000 at 15.27% gives an initial margin of about ₹40,000.
- Marking to market settles the day's profit or loss every evening in cash.
- A balance below the maintenance threshold triggers a margin call.
- Failing to meet it means being closed out, even if the price later recovers.

5 Leverage

PART B · MARGIN, LEVERAGE AND HEDGING · TOPIC 5 OF 9 · SERVES LEARNING OBJECTIVE 2

IN ONE LINE

Forty thousand rupees controls two lakh sixty-two thousand. A 5% move in the share becomes a 32% move in your deposit.

Why this matters

Leverage is why derivatives attract both hedgers and speculators, and why the same instrument can be prudent in one hand and ruinous in another.

The multiplier

Rahul controls ₹2,62,000 of exposure with a ₹40,000 deposit. Contract value divided by margin is **about 6.5**, so he has roughly 6.5 times leverage — a 1% move in the share price produces about a 6.5% move in the value of his deposit.

Both directions

If the futures price rises 5% to about ₹275, the gain is ₹13 per share, or ₹13,000 — roughly **+32%** on the ₹40,000 margin. If it falls 5% to about ₹249, the loss is the same ₹13,000, or roughly **−32%**.

A 5% market move produces a 32% swing in the deposit. Nearly a third of the capital gone in a single move — and a margin call may follow immediately.

Why the forced close is the real danger

Leverage magnifies profits when the market moves in the trader's favour and magnifies losses when it moves against. But the genuine hazard is not the magnified loss itself — it is the **forced close at an unfavourable moment**.

A trader with an unleveraged position can wait out an adverse move. A leveraged trader may be closed out before the recovery arrives, converting a temporary loss into a permanent one. That is what the day three and day four sequence in Topic 4 demonstrates.

A 5% MOVE IN THE SHARE IS A 32% MOVE IN THE DEPOSIT

	Futures price rises 5%	Futures price falls 5%
New price	About ₹275	About ₹249
Change per share	+₹13	−₹13
Total on 1,000 shares	+₹13,000	−₹13,000
As a share of the ₹40,000 margin	About **+32%**	About **−32%**
Consequence	A 5% move produces a 32% return	Nearly a third of the deposit gone; a call may follow

*Leverage = contract value ÷ margin = 2,62,000 ÷ 40,000 ≈ **6.5 times**. It does not change the size of the price move — it changes what proportion of your money it represents.*

Figure 5.1 Leverage, symmetrical in both directions. The danger lies in what happens next.

KEY IDEA

Leverage is not risk. It is an *amplifier* applied to whatever risk already exists — which is why the same 6.5 times leverage is prudent for a hedger reducing an exposure and dangerous for a speculator creating one.

EXAM POINTER

Compute leverage as contract value divided by margin, then apply it to a stated percentage move and express the result as a proportion of the deposit. The high-value point is the forced close — that leverage converts temporary adverse moves into permanent losses.

QUICK RECAP · TOPIC 5

- Leverage = contract value ÷ margin deposit; here 2,62,000 ÷ 40,000 ≈ 6.5 times.
- A 1% price move produces roughly a 6.5% move in the deposit.
- A 5% move produces about a 32% swing in the ₹40,000 margin, in either direction.
- Leverage amplifies existing risk rather than creating it.
- The real danger is the forced close, which turns a temporary loss into a permanent one.

6 Hedging a Portfolio with Index Futures

PART B · MARGIN, LEVERAGE AND HEDGING · TOPIC 6 OF 9 · SERVES LEARNING OBJECTIVE 3

IN ONE LINE

Shorting index futures switches off the market for a few weeks — and the number of contracts depends on beta, not just on portfolio value.

Why this matters

This is the practical payoff of the whole first half of the week, and it connects directly back to the beta estimated in Week 6.

The problem

Meera manages an equity fund worth ₹10 crore. She is not pessimistic long term and still believes in her holdings, but an uncertain budget week is approaching and a short sharp fall could wipe out months of gains. Selling the entire portfolio would be slow, costly and contrary to her investment view.

Instead she shorts Nifty index futures with a notional value of roughly ₹10 crore. If the market falls 5%, the portfolio loses about ₹50 lakh and the short futures gains about ₹50 lakh. If it rises 5%, the portfolio gains ₹50 lakh and the futures loses it. The two approximately offset.

The trade-off

A futures hedge is **symmetrical**. It removes the downside but equally removes the upside for the hedge period. Buying protective puts on the index instead would cost a premium but keep the upside intact — the difference between a free price lock and paid insurance, exactly as Topic 3 established.

Why beta changes the number

Saying Meera should short futures worth roughly the portfolio value hides an important factor. Beta measures how strongly a portfolio moves relative to the market: a portfolio with a beta of 1.2 is expected to move about 1.2 times as much, so if the market falls 1% the portfolio may fall about 1.2%.

Hedging only the rupee value would therefore leave part of the exposure uncovered. The correct calculation scales by beta:

Number of contracts = (portfolio value × beta) ÷ value of one futures contract

A practical note on lot sizes

The NSE and SEBI periodically revise index futures lot sizes to keep the notional value of one contract within a target range — recently around ₹15 lakh for a Nifty 50 contract. **Lot size should never be memorised as a permanent number**; always check the current contract specifications published by the exchange. The concept stays the same even when the contract size changes.

HOW MANY CONTRACTS? · BETA IS THE STEP EVERYONE SKIPS

Step	Working	Result
Portfolio value	Given	₹5 crore
Portfolio beta	Estimated by regression, as in Week 6	1.2
Value of one contract	Nifty at 24,000 × illustrative lot of 25 units	₹6,00,000
Value to be hedged	₹5 crore × 1.2	₹6 crore
Contracts required	₹6 crore ÷ ₹6 lakh	**100, taken short**
Ignoring beta	₹5 crore ÷ ₹6 lakh	About 83 — leaving roughly a sixth unprotected

Hedging only the rupee value gives 83 contracts against the 100 required, leaving ***about a sixth of the portfolio's market exposure uncovered*** *in a major decline.*

Figure 6.1 The hedge ratio. Beta is the difference between 83 contracts and 100.

REAL-WORLD EXAMPLE

The choice between futures and puts is not theoretical. A fund facing a single uncertain event with a strong long-term view will often prefer futures — cheap, temporary, nearly complete. A fund whose mandate requires it to capture upside will pay for puts instead, accepting the premium as a cost of staying invested.

EXAM POINTER

Write the hedge ratio formula, compute the value of one contract, and show both the beta-adjusted and unadjusted figures so the gap is visible. State that the hedge is symmetrical — it removes upside as well as downside — and note that lot sizes are revised periodically.

QUICK RECAP · TOPIC 6

- Shorting index futures offsets portfolio losses when the market falls.
- A futures hedge is symmetrical: it removes the upside along with the downside.
- Number of contracts = (portfolio value × beta) ÷ value of one contract.
- A ₹5 crore portfolio at beta 1.2 needs 100 contracts, not 83.
- Lot sizes are revised periodically by the exchange and should never be memorised.

PART C · OPTION STRATEGIES

Combining the building blocks

7 Moneyness

PART C · OPTION STRATEGIES · TOPIC 7 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

Same trader, same view, same underlying — and three strikes create three completely different positions.

Why this matters

Moneyness is the vocabulary that makes strategy selection possible. Every strategy in the next two topics deliberately chooses a strike based on it.

Three strikes, one view

Nifty sits at 24,000 on budget day. Arjun believes the budget will support growth and decides to buy a call — but which strike?

He could buy the **23,500 strike**, already below the market and more expensive because it carries intrinsic value. He could buy the **24,000 strike**, exactly at the market level. Or he could buy the cheaper **25,000 strike**, far above the market, which is inexpensive but needs a substantial rise before it gains any intrinsic value.

Same trader, same view, same underlying — three very different positions.

The definition

Moneyness describes the relationship between the spot price and the strike price. It tells us whether exercising immediately would produce a gain, and it determines how much of a premium is intrinsic value and how much is time value.

What each region is good for

At-the-money options carry no intrinsic value but are highly sensitive to new information and to changes in the underlying price — which is why straddles are built from them. **Out-of-the-money** options are cheapest, consisting entirely of time value, but need a larger move to become worth exercising — which is why strangles use them.

Meera approaches budget day differently from Arjun. She does not know the direction, only that the reaction will be large, so she *combines* options rather than buying one. She has moved from trading individual options to constructing strategies.

MONEYNESS · THE SAME THREE WORDS MEAN OPPOSITE THINGS FOR CALLS AND PUTS

Relationship	Call option	Put option
Spot > strike	In the money — positive intrinsic value	Out of the money — no intrinsic value
Spot = strike	At the money — premium is entirely time value	At the money — premium is entirely time value
Spot < strike	Out of the money — no intrinsic value	In the money — positive intrinsic value

*At the money options are the **most sensitive** to news and cost the most in time value. Out of the money options are the **cheapest** and need the largest move to pay.*

Figure 7.1 Moneyness. The middle row is the only one where calls and puts agree.

COMMON PITFALL

Applying the call definition to puts. Spot above strike means in the money for a call and out of the money for a put. Reverse them and every strategy question that follows will be answered backwards.

EXAM POINTER

Give the three-by-two table in full rather than defining the terms for calls alone. Then link moneyness to cost: at the money carries the most time value and the greatest sensitivity, out of the money is cheapest and needs the largest move.

QUICK RECAP · TOPIC 7

- Moneyness is the relationship between spot price and strike price.
- For a call, spot above strike is in the money; for a put it is out of the money.
- At the money, the premium is entirely time value for both.
- At-the-money options are most sensitive to new information.
- Out-of-the-money options are cheapest but need the largest move to pay.

8 Protective Put and Covered Call

PART C · OPTION STRATEGIES · TOPIC 8 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

Combine shares with an option and you can engineer a payoff that imitates a completely different instrument.

Why this matters

These two strategies are the clearest demonstration in the course that shares, calls and puts are building blocks rather than separate products.

Strategy one · the protective put

A trader buys 1,000 Zomato shares at ₹255 and wants to keep holding them, but is uncertain about upcoming results. She buys one put with a strike of ₹250 at a premium of ₹10 per share. Two payoff lines now combine.

Below the strike the combined payoff goes **flat** — each additional rupee lost on the shares is recovered through the put. The maximum loss is fixed at ₹15 per share, or ₹15,000, however far the price falls. Above the strike the put expires worthless and the only cost is the premium, so the upside continues. Break-even is the ₹255 purchase price plus the ₹10 insurance cost: **₹265**.

The engineering insight: the resulting payoff resembles that of a long call. Combining shares with a put has produced a cash flow pattern that imitates another instrument entirely.

Strategy two · the covered call

Now the objective changes from protection to income. A patient long-term investor owns 1,000 Zomato shares bought at ₹255, expects the price to stay relatively stable, and would like to earn something while waiting. He sells a call with a strike of ₹270 at a premium of ₹8 per share, receiving ₹8,000.

Why it is called covered: the investor already owns the shares, so if the call buyer exercises he can simply deliver them. The short call is *covered* by the owned shares.

The premium provides a small cushion against losses, but the gain is **capped at ₹23,000** once the price reaches ₹270. Every additional rupee above the strike is surrendered to the call buyer — even at ₹400, the investor must still sell at ₹270. The payoff resembles a short put: he has created a synthetic short put position.

The analogy is a landlord renting out an asset for regular income — earning while waiting, but accepting that a sudden substantial rise in value may not be fully enjoyed.

TWO STRATEGIES, TWO SHAPES · OWN THE SHARES IN BOTH CASES

Price on expiry	Protective put · combined	Covered call · combined	Note
₹230	−₹15 per share	−₹17 per share	Put has created a floor; call only cushions
₹247	−₹15 per share	Nil	Covered call break-even
₹250	−₹15 per share	+₹3 per share	Protective put floor reached
₹255	−₹10 per share	+₹8 per share	The purchase price
₹265	Nil	+₹18 per share	Protective put break-even
₹270	+₹5 per share	+₹23 per share	Covered call ceiling reached
₹300	+₹35 per share	+₹23 per share	Put keeps rising; call is capped
₹350	+₹85 per share	+₹23 per share	The gap widens indefinitely

*The protective put creates a **floor under losses** and resembles a long call. The covered call creates a **ceiling on gains** and resembles a short put.*

Figure 8.1 The two strategies compared at every price. Read the last three rows to see the trade-off.

KEY IDEA

Shares, calls and puts are three building blocks. Combined, they produce payoffs that imitate other instruments — which is the heart of **financial engineering**, and the reason a desk can manufacture almost any risk profile a client asks for.

EXAM POINTER

For each strategy give the construction, the premium direction, the objective, the effect on the payoff and what it resembles. Compute both break-evens: purchase price plus premium for the protective put, purchase price minus premium received for the covered call.

QUICK RECAP · TOPIC 8

- A protective put means owning shares and buying a put; it creates a floor and resembles a long call.
- Maximum loss is (purchase price − strike) + premium = ₹15 per share; break-even is ₹265.
- A covered call means owning shares and selling a call; it creates a ceiling and resembles a short put.
- Maximum gain is (strike − purchase price) + premium = ₹23 per share, reached at ₹270.
- Combining building blocks to imitate another instrument is financial engineering.

9 Straddle and Strangle

PART C · OPTION STRATEGIES · TOPIC 9 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

Two strategies that take a position on the size of a move without committing to a direction.

Why this matters

The closing topic of the derivatives pair. Both strategies exist because there are moments — results, budgets, election counts — when the magnitude of the reaction is far more predictable than its sign.

Why a directional view is not always available

Everything so far required knowing which way the price would go. But before earnings announcements, election results and important rate decisions, a trader may be confident a stock will move sharply and have no idea which way. Options allow a position on the *size* of a movement without committing to a direction.

The long straddle

Structure: at the same strike, buy one call and one put. With Zomato at ₹260 on the evening before results, a trader buys the ₹260 call at ₹12 and the ₹260 put at ₹11. Total premium ₹23 per share, or **₹23,000** for the lot. Straddles are built from *at-the-money* options, which are the most sensitive to unexpected moves.

The shape is a letter **V**. Its lowest point sits at the strike of ₹260, where both options expire worthless and the maximum loss of ₹23,000 occurs. Break-evens are the strike minus the total premium (**₹237**) and the strike plus it (**₹283**).

The long strangle

The straddle is powerful but expensive, because both options are at the money. The strangle is the cheaper alternative: again a call and a put, but at *different* strikes, both bought out of the money.

Structure: buy the ₹280 call at ₹6 and the ₹240 put at ₹5. Total premium ₹11 per share, or **₹11,000** — less than half the straddle. Lower break-even 240 − 11 = **₹229**; upper break-even 280 + 11 = **₹291**.

The payoff differs from the sharp V: between the two strikes it stays flat at the maximum loss, producing a **flat-bottomed valley**. Profit begins only below ₹229 or above ₹291.

Choosing between them

Moneyness explains the cost difference entirely. The straddle uses expensive at-the-money options and therefore responds to a smaller move; the strangle uses cheap out-of-the-money options and needs a larger one. If the concern is the whole market rather than one stock, both should be built on index options such as Nifty options rather than a single share.

STRADDLE AND STRANGLE · SAME IDEA, DIFFERENT COST AND DIFFERENT SHAPE

	Long straddle	Long strangle
Strikes	One strike, at the money — both at ₹260	Two strikes, spread apart — ₹280 call, ₹240 put
Moneyness used	At the money — more expensive	Out of the money — cheaper
Total premium	₹12 + ₹11 = ₹23,000	₹6 + ₹5 = ₹11,000
Payoff shape	A sharp V	A flat-bottomed valley
Break-evens	₹237 and ₹283	₹229 and ₹291
Maximum loss	₹23,000, at exactly ₹260	₹11,000, anywhere between the strikes
Movement needed	Benefits from a smaller move	Requires a larger move
Suits a trader who	Wants the position to respond quickly	Expects a very large move and wants lower cost

*At ₹210 the straddle nets **+₹27,000**; at ₹320, **+₹37,000**. The strangle costs less than half as much to put on, but does not start paying until ₹229 or ₹291.*

Figure 9.1 The two volatility strategies. Cheaper entry buys a wider dead zone.

KEY IDEA

Both strategies are **bets on volatility, not direction**. The trader is indifferent to which way the market goes and cares only that it goes far enough — which is a genuinely different kind of view from anything earlier in the course.

EXAM POINTER

Give the construction and both break-evens for each. The formula to state: straddle break-evens are strike ± total premium; strangle break-evens are put strike minus total premium and call strike plus total premium. Describe both shapes — sharp V against flat-bottomed valley — and explain the cost difference by moneyness.

QUICK RECAP · TOPIC 9

- Both strategies take a position on the size of a move, not its direction.
- A straddle buys a call and a put at the same at-the-money strike; premium ₹23,000 here.
- Straddle break-evens are strike ± total premium: ₹237 and ₹283.
- A strangle uses two out-of-the-money strikes and costs less than half as much.
- Strangle break-evens are ₹229 and ₹291, with a flat-bottomed valley between the strikes.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain how a forward becomes a futures contract and compare futures with options	Topics 1, 2 and 3 · Figures 1.1, 2.1, 3.1
2	Describe margin and daily settlement, and compute leverage	Topics 4 and 5 · Figures 4.1, 5.1
3	Compute the number of contracts needed to hedge a portfolio	Topic 6 · Figure 6.1
4	Apply moneyness and construct four option strategies	Topics 7, 8 and 9 · Figures 7.1, 8.1, 9.1

The module in six sentences

A right must be purchased, but two promises exchanged cost nothing at the outset — which is why an option carries a premium and a futures contract does not, and why only one of them can be abandoned. That single asymmetry produces a futures payoff that is one straight line with no kink, no floor and a break-even at exactly the locked price. Because both parties are bound, the exchange requires margin from each and settles profits and losses in cash every evening, so a balance below the maintenance threshold triggers a call that can force a close before any recovery arrives. That same margin creates leverage of about 6.5 times, turning a 5% price move into a 32% swing in the deposit. Hedging a portfolio requires scaling by beta, so a ₹5 crore fund with a beta of 1.2 needs 100 contracts rather than the 83 its rupee value alone would suggest. And combining shares with options engineers new payoffs entirely — a floor from the protective put, a ceiling from the covered call, and in the straddle and strangle a position on the size of a move rather than its direction.

HOW THE NINE TOPICS CONNECT

PART A · THE BINDING PROMISE

1 · Forward to futures
2 · The payoff
3 · Options versus futures

TAKEAWAY
No premium, no floor

PART B · MACHINERY

4 · Margin
5 · Leverage
6 · Hedging

TAKEAWAY
Promises need enforcement

PART C · STRATEGIES

7 · Moneyness
8 · Put and call strategies
9 · Straddle and strangle

TAKEAWAY
Blocks, not products

A right must be purchased; a promise exchanged for another promise costs nothing. *That asymmetry explains the premium, the payoff shape, the margin and the risk of every instrument this week.*

Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check

Term	Definition
Forward contract	A private, customised agreement to trade at a set price on a future date.
Futures contract	A standardised forward supported and guaranteed by an exchange.
Counterparty risk	The possibility that the other party fails to honour its promise.
Basis	The gap between spot and futures price, which converges to zero at expiry.
Long position	The side promising to buy; gains when the underlying rises.
Short position	The side promising to sell; gains when the underlying falls.
Margin	A refundable security deposit required of both parties to a futures contract.
Marking to market	Daily settlement of gains and losses into the margin account.
Margin call	A demand for additional funds when the balance falls below the required minimum.
Leverage	Contract value relative to margin deposited; magnifies gains and losses alike.
Index futures	Futures on the level of an index, settled in cash rather than by delivery.
Hedge ratio	The number of contracts needed to hedge a portfolio, adjusted for beta.
Moneyness	The relationship between spot and strike that determines intrinsic value.
In the money	Exercising immediately would produce a gain.
At the money	Spot equals strike; the premium is entirely time value.
Out of the money	Exercising immediately would not make sense.
Protective put	Owning shares and buying a put; creates a floor under losses.
Covered call	Owning shares and selling a call; creates a ceiling on gains.
Synthetic position	A payoff engineered from other instruments to imitate a different one.
Long straddle	A call and a put at the same at-the-money strike; a bet on size, not direction.
Long strangle	A call and a put at different out-of-the-money strikes; cheaper, needs a bigger move.
Financial engineering	Creating a desired cash flow pattern by combining financial instruments.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 Why does a futures contract require no premium when an option does?

> **Because the agreement is symmetrical.** Both parties are equally bound, so neither has received a one-sided right the other must pay for. An option buyer receives something the writer does not — the ability to walk away — and the premium is the price of that.

Q2 A trader is long one futures contract at ₹262 with a lot of 1,000. Where is break-even, and why does the payoff have no kink?

> **Break-even is ₹262 — the locked price, with nothing to add.** There is no kink because there is no right to abandon: the contract binds at every price, so nothing causes the line to change direction. There is also no floor under the loss.

Q3 Initial margin is ₹40,000 on a contract worth ₹2,62,000. The price moves 5%. What happens to the deposit?

> **It swings about 32%.** Leverage = 2,62,000 ÷ 40,000 ≈ 6.5. A 5% move is ₹13 per share, or ₹13,000, which is roughly a third of the ₹40,000 deposit — in either direction.

Q4 A ₹5 crore portfolio has a beta of 1.2. Nifty is at 24,000 with a lot of 25 units. How many contracts are needed, and what happens if beta is ignored?

> **100 contracts short.** Contract value = 24,000 × 25 = ₹6,00,000. Value to hedge = ₹5 crore × 1.2 = ₹6 crore. ₹6 crore ÷ ₹6 lakh = 100. Ignoring beta gives 83 contracts, leaving about a sixth of the market exposure unprotected.

Q5 An investor owns shares at ₹255 and sells a ₹270 call for ₹8. The share reaches ₹400. What does he earn?

> **₹23 per share — and no more.** The gain is capped at (270 − 255) + 8 = ₹23 once the price passes ₹270. Every rupee above the strike goes to the call buyer. This is a covered call, and the ceiling is the price of the income.

Q6 A trader buys a ₹260 call at ₹12 and a ₹260 put at ₹11 before results. Where are the break-evens, and what view does this express?

> **₹237 and ₹283 — a view on size, not direction.** Total premium is ₹23, so break-evens are strike ± premium. This is a long straddle: the trader profits if the price moves far enough either way, and loses the full ₹23,000 only if it finishes exactly at ₹260.